Assignment week 08

Preamble

A **firewall** is a way to regulate the incoming and outgoing traffic on your network. A well-configured firewall is crucial for the security of servers.

The default firewall configuration tool for mint is ufw.

Ufw was developed to ease iptables firewall configuration and provides an easy way to create an IPv4/IPv6 host-based firewall.

ufw by default is initially disabled. It is not intended to provide complete firewall functionality via its command interface, but instead provides an easy way to add or remove simple rules. It is currently mainly used for host-based firewalls.
First, ufw needs to be enabled. From a terminal prompt enter:

sudo ufw enable

To open a port (SSH in this example):

sudo ufw allow 22

To close an opened port:

sudo ufw deny 22

To remove a rule, use delete followed by the rule:

sudo ufw delete deny 22

To see the firewall status, enter:

sudo ufw status

For more verbose status information use:

sudo ufw status verbose

ufw can be disabled by:

sudo ufw disable

You can rest the ufw to initial settings with:

sudo ufw reset

There is also a graphical user interface version of ufw – called gufw. Install gufw by using the Advanced package Tool (APT) as follows:

sudo apt install gufw -y

The assignment

Do the following:

1. Enable the UFW
2. Use the UFW to block traffic to the SSH service
3. Use the UFW to block traffic to a webserver (port 80)
4. Allow traffic to the webserver and continue to block SSH traffic
5. Review a gufw website: https://itsfoss.com/set-up-firewall-gufw/
6. Install gufw using the Advanced Package Tool
7. Run gufw – simply type:

   gufw &

Take screenshots to show that you have done each of the above tasks

8. You are a gamer, you also like to share content with friends when they visit your house, you like to travel a lot with your Linux laptop. You also use the device for work. How might you configure the firewall on your device to keep your device and data safe? Illustrate your answer with screenshots of the sort of rules you would set up.

9. "A device with a poorly configured firewall may end up appearing on Shodan". Give a brief discussion as to why this might be the case.

Submit your work in a pdf document to the VLE